

YUN-RONG (LAUREN) LUO

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RESEARCH INTEREST

Distributed systems, formal verification, and programming languages

EDUCATION

University of Michigan, Ann Arbor

Ph.D. Candidate in Computer Science and Engineering

• Research Advisor: Prof. Manos Kapritsos

Ann Arbor, MI

08/2023-Present

National Taiwan University (NTU)

M.S. in Electronics Engineering

• Research Advisor: Prof. Jie-Hong Roland Jiang

Taipei, Taiwan

09/2021-2023

B.S. in Electrical Engineering

09/2017-06/2021

PUBLICATIONS

Conference

- Yun-Rong Luo**, Aman Goel, Karem Sakallah, "QSM-Cutoff: Systematic Derivation of Quantified Cutoff Formulas for Distributed Protocols.", 2025 International Conference on Computer Aided Verification (CAV), 2025 [\[link\]](#)
- Che Cheng*, **Yun-Rong Luo***, Jie-Hong Roland Jiang, "Knowledge Compilation for Incremental and Checkable Stochastic Boolean Satisfiability." 2024 International Joint Conference on Artificial Intelligence (IJCAI), 2024 [\[link\]](#)
- Yu-Neng Wang*, **Yun-Rong Luo***, Po-Chun Chien*, Ping-Lun Wang, Hao-Ren Wang, Wan-Hsuan Lin, Jie-Hong Roland Jiang, Chung-Yang Ric Huang, "Compatible Equivalence Checking of X-Valued Circuits." 2021 IEEE/ACM International Conference On Computer Aided Design (ICCAD), 2021 [\[link\]](#)

Journal

- Yun-Rong Luo***, Che Cheng*, Jie-Hong Roland Jiang, "A Resolution Proof System for Dependency Stochastic Boolean Satisfiability." Journal of Automated Reasoning, 2023 [\[link\]](#)

RESEARCH EXPERIENCE

Graduate Student Research Assistant (UMich)

Ann Arbor, MI

- Investigating performance guarantees for distributed systems via formal verification.
- Developed [QSM-Cutoff](#): automated inductive invariant synthesis for distributed protocols via logic minimization.

Graduate Student Research Assistant (NTU)

Taipei, Taiwan

- Extended a certified model-counting toolchain to [cert-SSAT](#) for stochastic Boolean satisfiability (SSAT) solver certification.
- Developed a resolution proof system for dependency stochastic Boolean satisfiability (DSSAT).

Undergraduate Student Researcher (NTU)

Taipei, Taiwan

- Contributed to the 1st-place-winning strategy for *equivalence checking of X-valued circuits* at the 2020 ICCAD CAD Contest.

WORK EXPERIENCE

Cadence Design Systems

Hsinchu, Taiwan

R&D Intern, Formal Verification Team

07/2020-09/2020

- Generated *engineering change order (ECO)* test cases for the Conformal ECO tool.

TEACHING EXPERIENCE

EECS 376: Foundations of Computer Science (UMich)

Graduate Student Instructor

05/2025-06/2025

EECS 270: Introduction to Logic Design (UMich)

Graduate Student Instructor

08/2024-04/2025

Logic Synthesis and Verification (NTU)

Teaching Assistant

09/2021-01/2022

HONORS AND AWARDS

2021 Research Creativity Award, Ministry of Science and Technology of Taiwan (awarded to 200/3000+ research works)

2020 1st Place, Problem A, 2020 CAD Contest at ICCAD (186 teams competed in 3 problems)

SKILLS

Programming Languages C/C++, Python, Go, OCaml, Verilog

Tools Shell scripting, Git, $\text{\LaTeX}$

Verification Tools [Dafny](#), SAT/SMT solvers,